

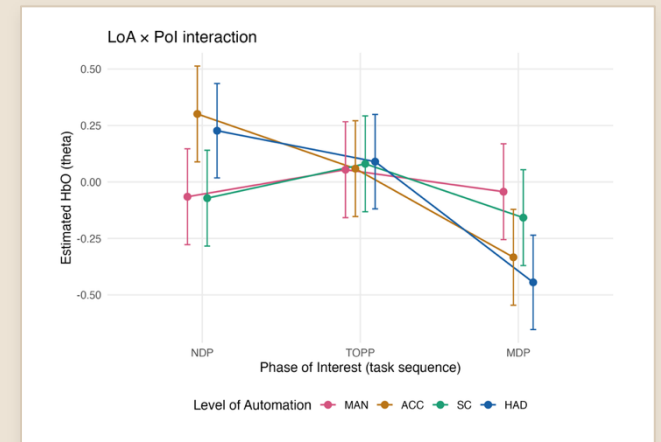
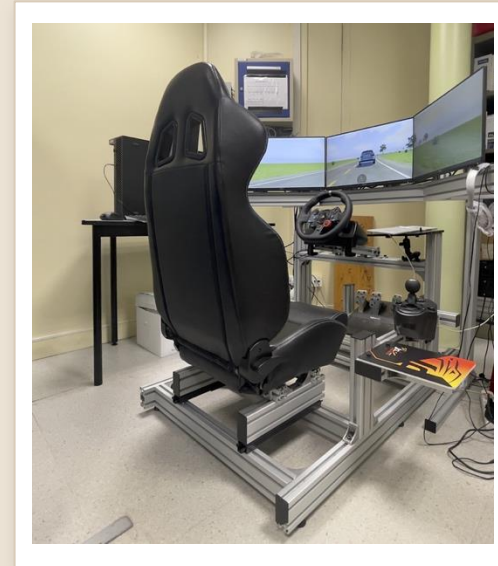
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RESEARCH ARTICLE · MASTER'S INTERNSHIP · 2026

Higher automation, weaker driver re-engagement after takeover

A driving simulator and brain-imaging (fNIRS) study of how the level of automation shapes cortical activity before, during and after a take-over.



Study at a glance

The question. Automated cars still hand control back to the driver. I studied how the brain’s cognitive systems behave across this handover, and whether deeper automation makes the return harder.

The answer. After takeover, activity in the cognitive control region was lower than before, and the more automation had preceded, the lower it was.

Context	Master's professional internship, EMC Lab (Laboratoire d'Étude des Mécanismes Cognitifs), Université Lumière Lyon 2
Supervisor	Jordan Navarro
My role	Co-defined the question and method with my supervisor; recruited and tested 71 volunteers; analysed all fNIRS data; wrote the thesis. Built on an existing simulator and scenarios (Deniel et al., 2024).
Duration	6 months, January to June 2026
Participants	34 drivers analysed (71 tested; 37 datasets abandoned due to technical issues)
Methods	Driving simulator, fNIRS (35 channels, 4 regions), GLM, linear mixed-effects models
Tools	Python (MNE-NIRS), R (lme4, emmeans), Cortiview

Four questions structure this study

01

WHY

Why does it matter?

Automated cars still hand control back, yet we do not know how the brain's control systems recover.

02

HOW

How did I test it?

34 drivers, 4 automation levels, 3 phases of the take-over cycle, fNIRS over 4 functional regions.

03

RESULTS

What did we learn?

Activity dropped after takeover, and cognitive control re-engaged less the more automation had preceded.

04

NEXT

What comes next?

Design implications, five limitations, and better ways to model the take-over cycle.

Automated cars still hand control back to the driver

THE NEAR FUTURE

Highly automated driving

At SAE Levels 3–4 (the industry's 0–5 automation scale), the car handles the whole driving task within its design limits, but the driver must take over on request. Full autonomy (Level 5) remains years away (Navarro, 2019).

THE WEAK POINT

The take-over request (TOR)

A TOR is the system's signal that the driver must resume control. Resuming control from automation is not a simple handoff but a demanding cognitive transition (Hoc et al., 2009).

WHY THE HANDOVER IS HARD

Drivers who only supervise drift out of the loop: they sample the road less (Deniel et al., 2024), lose vigilance, and develop passive fatigue (Saxby et al., 2013). Even moving the hands back to the wheel adds about 1.33 s to take-over time (Zhang et al., 2019).

Every car that can hand back control assumes the driver can come back. The open question is how fully, and how fast.

Four functions, studied apart, used together

● Monitoring

Checking the task over time to keep it on track (Stuss & Alexander, 2007). It feeds situation awareness, the driver's up-to-date picture of the road (Endsley, 2001).

● Task switching

Reconfiguring from one task set to another, with a measurable switch cost (Monsell, 2003). Here: from passive supervisor to active driver.

● Cognitive control

Top-down regulation of goal-directed behavior, maintained by the prefrontal cortex (Miller & Cohen, 2001; Friedman & Robbins, 2022).

● Motor control

Eye–hand coordination for steering. Active motor engagement organises the whole perceptual-cognitive loop (Deniel et al., 2024).

THE RESEARCH GAP

Each function has been mapped mostly in isolation, which hides how they work together in real tasks (Benn et al., 2014). How the four are integrated across a complete take-over cycle remains poorly understood.

What the study set out to answer

How does the level of automation shape cortical activity across the three phases of the take-over cycle?

NDP

Nominal driving

Before the TOR, automation is active.

PREDICTION

Disengagement

Higher automation lowers activity in motor, monitoring and cognitive control regions.



TOPP

Take-over preparation

The 7 s after the TOR.

PREDICTION

Switch cost

The rise in switching and monitoring activity from NDP grows with automation.



MDP

Manual driving

The seconds after manual control resumes.

PREDICTION

Costly re-engagement

Motor and cognitive control activity is amplified, more so after higher automation.

Overall hypothesis: each effect, including the change between phases, scales linearly with the level of automation.

Four automation levels, three phases, one scenario

INDEPENDENT VARIABLE 1 · LEVEL OF AUTOMATION (LOA)

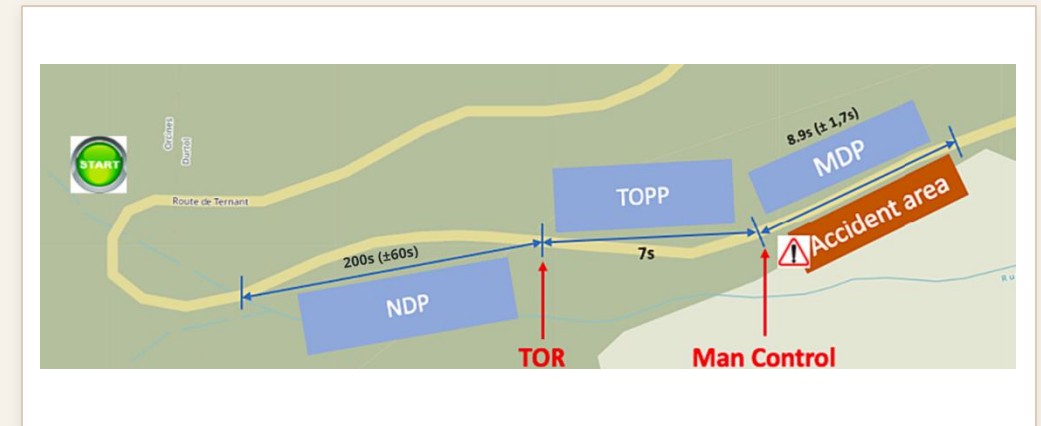
Level	SAE	Speed	Steering
MAN · manual	0	Driver	Driver
ACC · adaptive cruise	1	System	Driver
SC · shared control	2	System	Shared (haptic)
HAD · highly automated	3	System	System, hands off

Shared control: the driver steers, and the wheel pushes back proportionally when the car drifts from its lane. Within-subject design, 4 counterbalanced orders.

DEPENDENT VARIABLE

θ , the GLM activation coefficient of oxygenated hemoglobin (HbO): a larger θ means a stronger task-evoked response, which suggests greater neural demand.

INDEPENDENT VARIABLE 2 · PHASE OF INTEREST (POI)



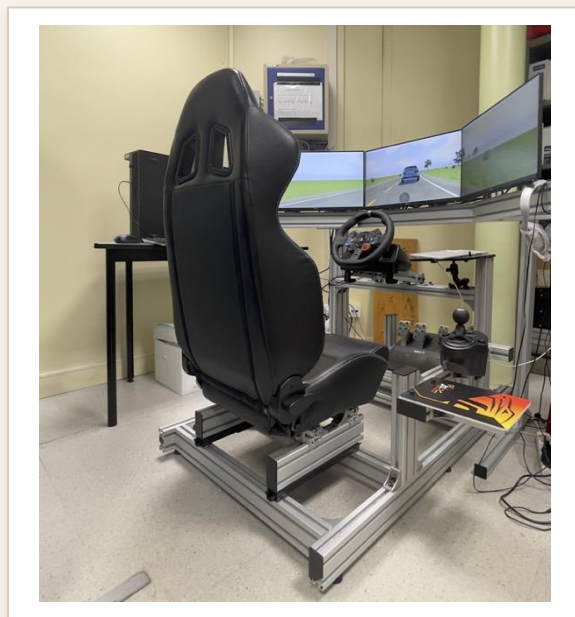
NDP nominal driving under automation (about 200 s)

TOPP 7 s visual countdown after the TOR, ending with “Off”

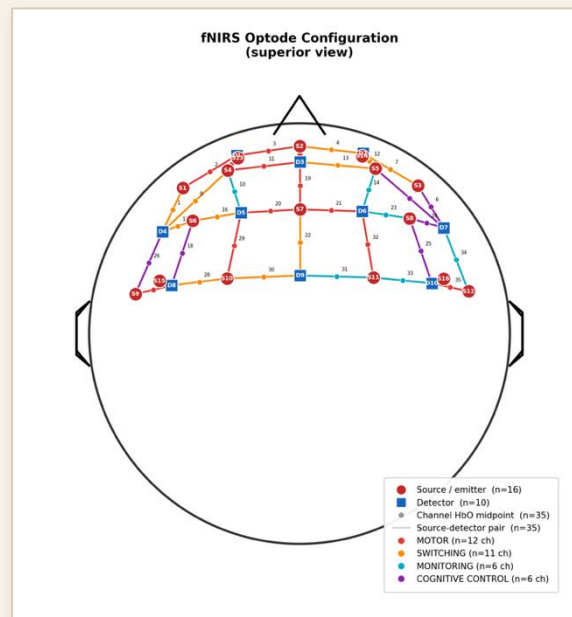
MDP manual driving to avoid an obstacle (truck, roadworks, parked cars, exit ramp)

5 road environments (rural, 2 highways, urban, mountain) × 4 automation levels per participant.

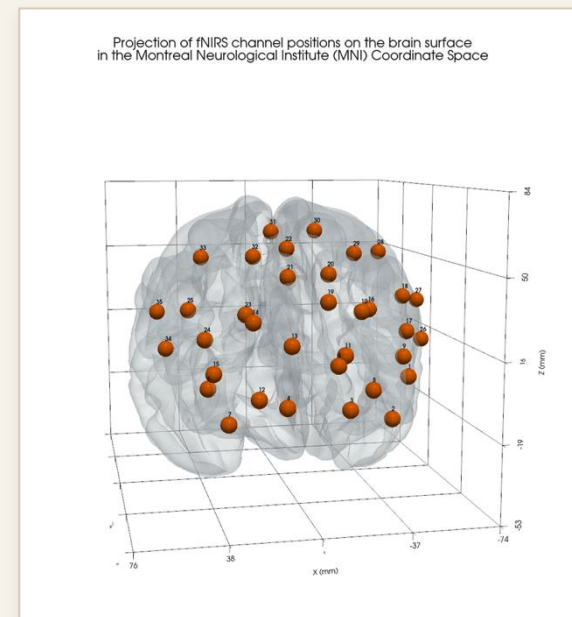
A driving simulator and a 35-channel fNIRS cap



Fixed-base simulator



Optode layout, coloured by region



Channel positions on a standard brain

PARTICIPANTS

34 drivers analysed: 29 women, 5 men, aged 20–45 (M = 24.2), licensed for 5 years on average. 71 were tested, but 37 datasets were lost to technical issues.

APPARATUS

Three screens (137.5° field of view), Logitech G29 wheel and pedals. Photon Cap C20 fNIRS: 16 sources, 10 detectors.

fNIRS (functional near-infrared spectroscopy) measures blood-oxygenation changes in the outer cortex using light. Here, 35 long channels (≈ 30 mm) were grouped into four regions: motor (12), switching (11), monitoring (6) and cognitive control (6). Four short channels (≈ 10 mm) captured scalp and systemic signals to be regressed out.

From raw light signals to hypothesis tests

I · Preprocessing (Python, MNE-NIRS)

Wavelet motion correction, 0.01–0.1 Hz band-pass filter, modified Beer–Lambert law; analyses on the 35 HbO channels.

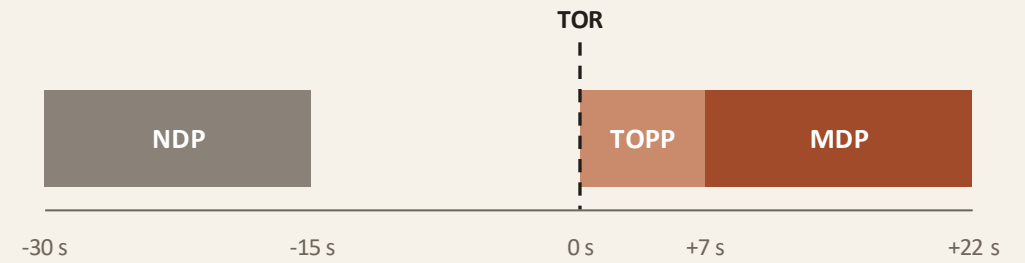
II · First level: GLM per channel

One regressor per LoA × PoI (4 × 3), convolved with a canonical hemodynamic response function, gives θ per condition.

III · Group level: mixed model (R, lme4, emmeans)

$\theta \sim \text{ROI} \times \text{LoA} \times \text{PoI} + (1 \mid \text{participant})$. Orthogonal polynomial contrasts over ordered LoA (MAN < ACC < SC < HAD), Tukey pairwise tests, and phase differences crossed with the linear LoA trend.

ANALYSIS WINDOWS, RELATIVE TO THE TOR



NDP: 30 to 15 s before the TOR. TOPP: the 7 s countdown. MDP: the 15 s after manual control resumes.

$$\theta \sim \text{ROI} \times \text{LoA} \times \text{PoI} + (1 \mid \text{participant})$$

Linear mixed-effects model (LMM): fixed effects for region, automation and phase with all interactions, plus a random intercept per participant.

Four analytic choices, explained

01 Why short-separation channels?

Channels 10 mm apart mostly capture scalp blood flow and systemic signals, such as heartbeat and breathing. Regressing them out leaves a cleaner cortical signal in the 30 mm channels.

02 Why polynomial contrasts?

The hypotheses predict a dose-response: activity changing steadily as automation increases. A linear contrast tests exactly this, while quadratic and cubic terms flag departures from a straight line.

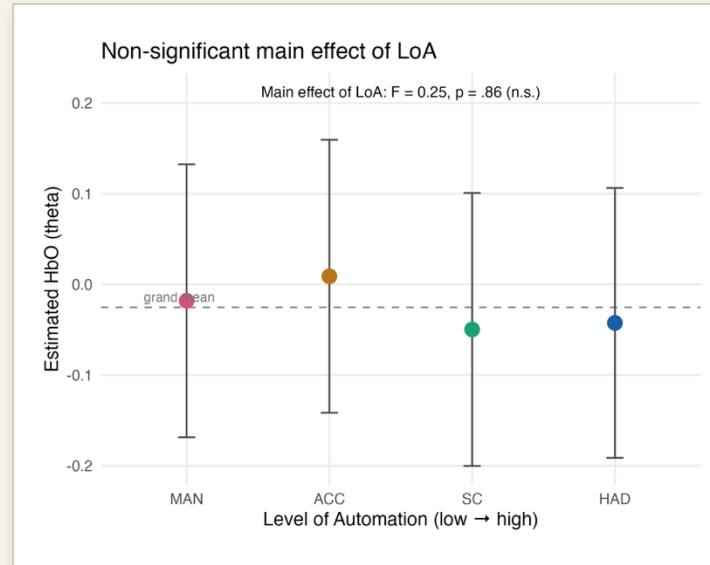
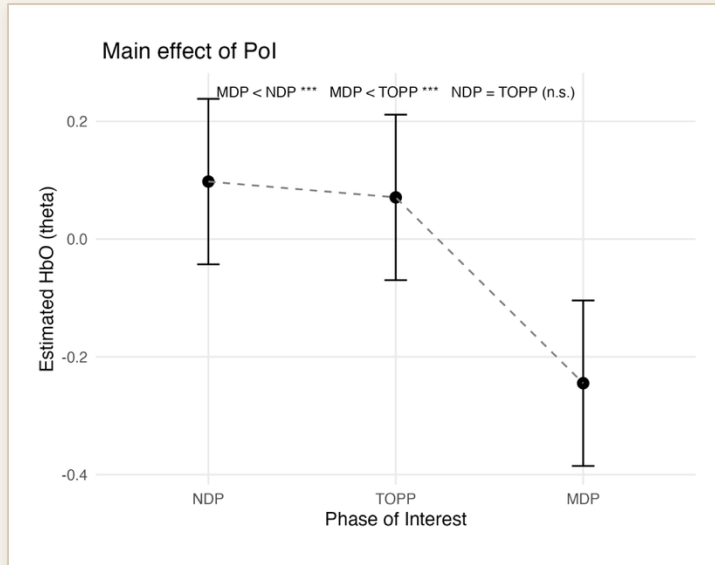
03 Why a random intercept only?

A model with per-participant random slopes for phase gave a singular fit, meaning it could not be estimated reliably. The simpler structure was kept, a transparent trade-off.

04 Why interaction contrasts?

The key predictions concern change between phases. Crossing each phase difference (e.g. MDP – NDP) with the linear automation trend tests whether the transition itself scales with automation.

Brain activity dropped after takeover



Phase main effect $F(2, 1231) = 17.18, p < .001$

MDP vs NDP $\Delta = -0.34 [-0.50, -0.19]$

MDP vs TOPP $\Delta = -0.32 [-0.47, -0.16]$

NDP vs TOPP $p = .91$

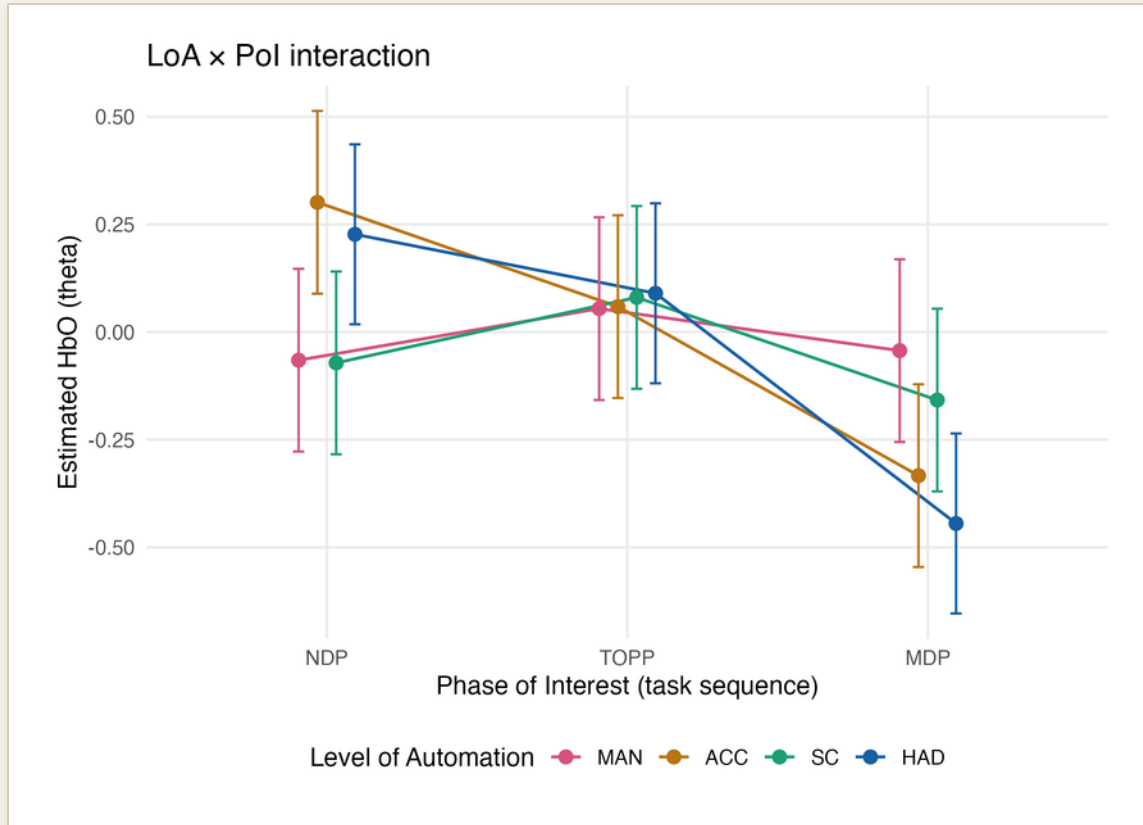
Automation main effect $F = 0.25, p = .86$

WHAT IT MEANS

The phase of the take-over cycle, not the automation level on its own, was the main source of variation. The drop was specific to manual driving after takeover.

Left: ϑ by phase. Right: ϑ by automation level. Bars are 95% confidence intervals.

Automation mattered, but only in certain phases



Automation × phase

 $F(6, 1231) = 4.03, p < .001$

NDP: ACC vs MAN and vs SC

 $\Delta = 0.37, p < .05$

MDP: MAN vs HAD

 $\Delta = 0.40, p < .05$

MDP – NDP under ACC

 $\Delta = -0.63, p < .0001$

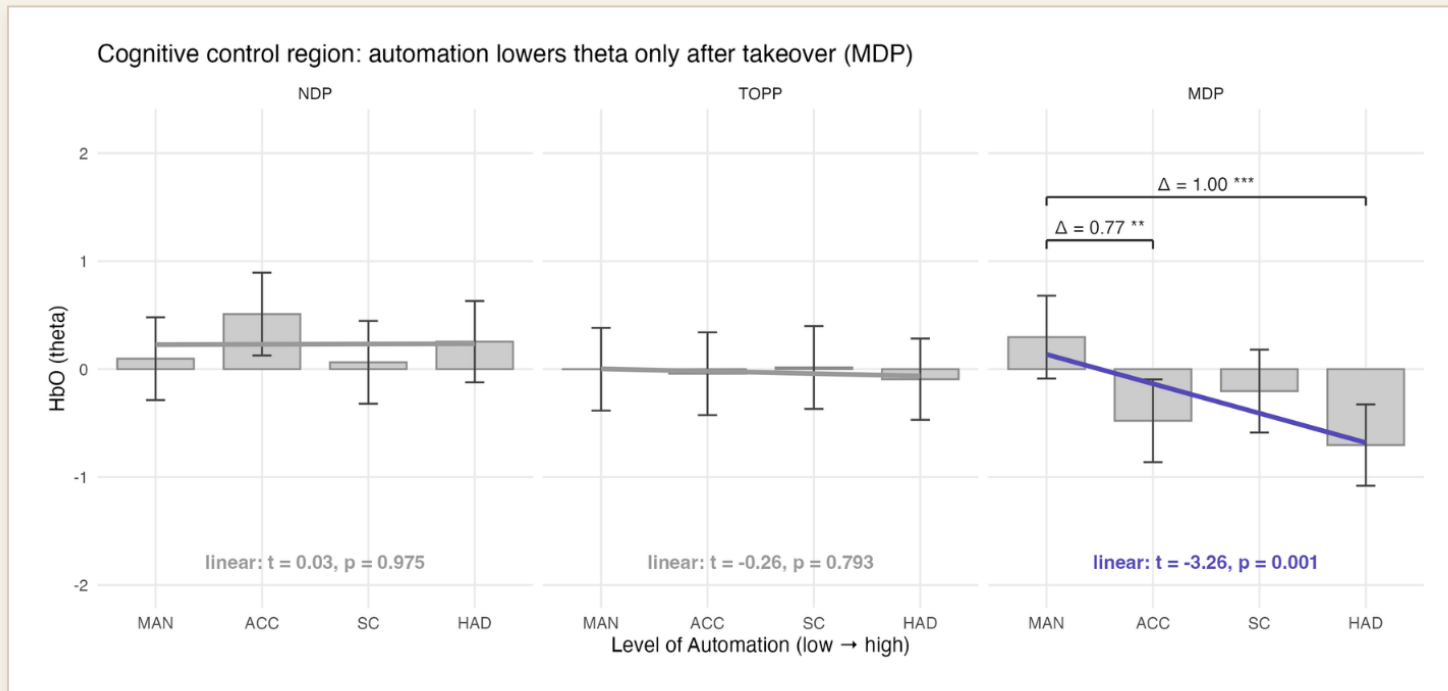
MDP – NDP under HAD

 $\Delta = -0.67, p < .001$

WHAT IT MEANS

Automation did not shift activity uniformly. After takeover, activity still reflected the automation that had preceded it: lower after highly automated driving than after manual driving.

More automation before, less cognitive control after



Linear trend in MDP

 $t(1282) = -3.26, p = .001$

MAN vs ACC in MDP

 $\Delta = 0.77, p = .02$

MAN vs HAD in MDP

 $\Delta = 1.00, p < .01$

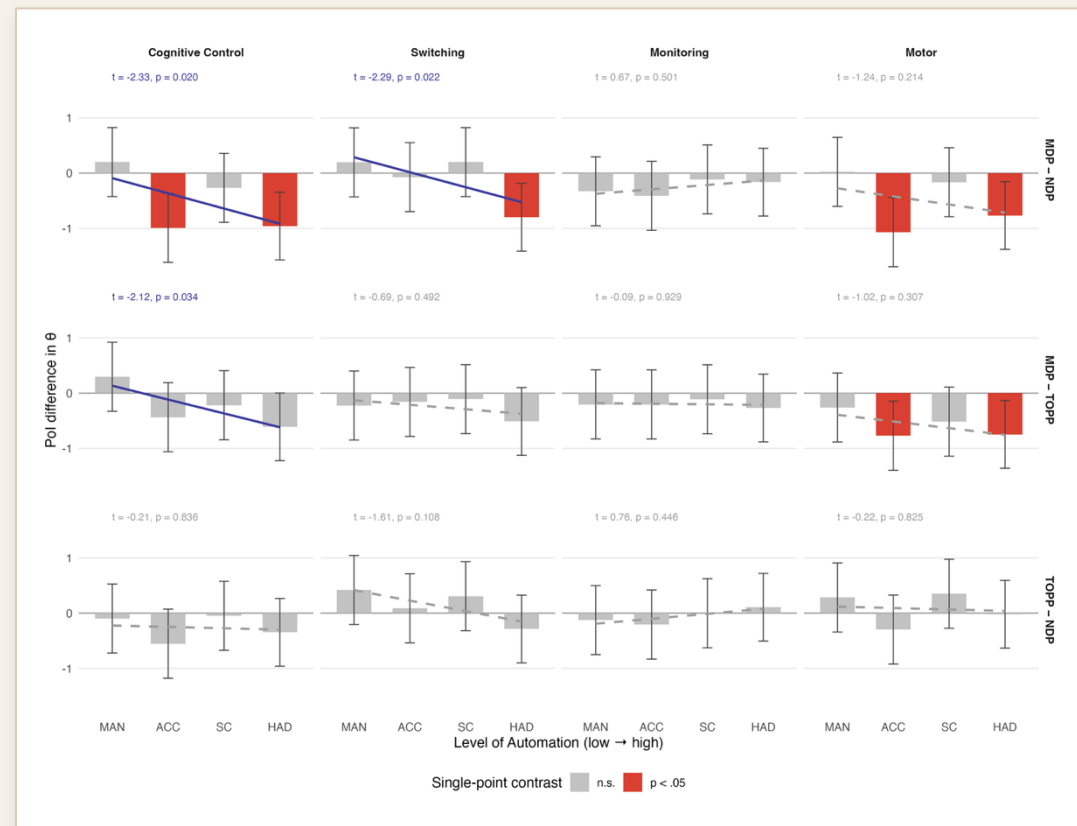
Trends in NDP and TOPP

all $p > .17$ MDP – NDP gap $\times$ LoA $t = -2.33, p = .02$ MDP – TOPP gap $\times$ LoA $t = -2.12, p = .03$

ϑ in the cognitive control region by automation level, for each phase. Purple line: significant linear trend.

Against the hypothesis. We predicted more effortful re-engagement (higher activity) after higher automation. We observed weaker re-engagement (lower activity), only after takeover.

Each region told a different story



Phase differences in θ by region (columns) and comparison (rows). Red: significant contrast.

- Cognitive control** Larger drop after takeover as automation increased (see Finding 03).
- Switching** An attenuated echo of cognitive control: MDP < NDP only under HAD ($\Delta = -0.80$); the MDP – NDP gap grew with automation ($t = -2.29$, $p = .022$); the within-MDP trend was marginal ($p = .057$).
- Motor control** Lower after takeover under ACC and HAD (MDP – NDP $\Delta = -1.07$ and -0.77) but insensitive to automation level ($ps > .21$). HbO was negative even in manual driving, an unexplained direction that may reflect artefacts.
- Monitoring** No reliable effect in any phase or comparison.

Not a cost of recovery, but a cost of disengagement

HYPOTHESIS SCORECARD

NDP · disengagement Automation lowers activity before the TOR	Not supported
TOPP · switch cost NDP → TOPP rise grows with automation	Not supported
MDP · re-engagement Activity amplified after higher automation	Reversed
Transitions Phase changes scale with automation	Partial

EXPECTED VS OBSERVED

Cost of recovery

Expected. Coming back after automation demands extra effort, so activity rises after takeover.

Cost of disengagement

Observed. Under-engagement persists after takeover, beyond the 7 s preparation, and deepens with prior automation.

INTERPRETATION

This extends the out-of-the-loop account (Endsley, 1995) beyond the takeover itself: re-engaging top-down control looks like a graded, incomplete process rather than a switch. Without behavioral data, this remains a neural interpretation.

What this could mean for automated driving

Design hypotheses from a simulator study without behavioral measures. They need behavioral validation before guiding design.

01

A takeover does not end when the hands are back

Cognitive control was still under-engaged after the 7 s preparation window. Support during the first seconds of manual driving, such as keeping alerts or light assistance active, may deserve testing.

02

Depth of automation sets the cost of coming back

The deeper the preceding automation, the weaker the re-engagement. The comfort of higher automation should be weighed against this lingering cost, for example by keeping drivers partly in the loop.

03

Evaluate takeovers beyond the takeover moment

Take-over time alone may miss a lingering under-engaged state. Evaluation protocols could track the following seconds, combining driving behavior, gaze and, where possible, neural measures.

Five limitations, and how to address them

LIMITATION

NEXT STEP

- | | |
|---|--|
| 1 No behavioral data The neural under-engagement cannot yet be linked to take-over performance or situation awareness. | → Record take-over time, steering and braking quality, and gaze. |
| 2 Small, imbalanced sample 37 of 71 datasets were lost; 29 of 34 participants were women. Power is limited, which may explain the marginal switching trend and null monitoring result. | → Recruit a larger, sex-balanced sample. |
| 3 Overlapping time windows TOPP and MDP are adjacent and the HbO response is slow, so the GLM cannot fully separate them. | → Test alternative models of the cycle (next slide). |
| 4 Surface-only measurement fNIRS reaches about 2–3 cm into the cortex; functional regions only partly capture deeper structures. | → Treat region labels as approximate cortical correlates. |
| 5 Fixed-base simulator No vestibular cues, real risk or real stakes, all of which may affect engagement during a genuine takeover. | → Validate in more realistic settings. |

Better ways to model the take-over cycle



Current model

Three fixed, disjoint blocks (NDP, TOPP, MDP), each estimated separately.



Option 1 · events

Model the TOR and the return to manual control as brief events, test the contrasts TOR – NDP and MDP – NDP, then compare these changes across automation levels with a difference-in-differences test at group level.



Option 2 · one block

Model one continuous block from the TOR to the end of the trial, and compare each automation level against manual driving.

Combined with behavioral take-over measures and a larger sample, these models would show whether the cognitive control deficit predicts real take-over quality.